Towards field-deployable, man-portable, infrared target detection of drones and vehicles using transformer-based computer vision models and embedded GPUs

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***Abstract*—Drones and vehicles can be used for safe or harmful purposes. Thus detecting them is crucial to ensure the safety of public spaces such as airports, stadiums, and campuses. Infrared cameras combined with computer vision are effective methods to provide surveillance because they can operate in day or night, and offer greater specificity than systems based on other modalities such as radio frequency communications or sound. When infrared cameras and computer vision models are combined with embedded computers and batteries, field deployable systems can be made and deployed remotely. Previous work has typically focused on tracking, not detection; use large computer vision models which cannot run on embedded computers; or have limited replicability for data collection. In this work, we address these shortcomings by prototyping detection of drones and vehicles using transformer-based computer vision models run on desktop and embedded GPUs on data collected from a commercially available infrared camera. (The camera is available for other labs to use.) Results are promising but further study such as hyperparameter tuning of models and hardware is needed.**

Keywords—computer vision, transformer-based computer vision models, infrared camera, desktop and embedded GPU, drone detection, vehicle detection

I. Introduction

Drones are increasingly common and are used for many tasks like photography or delivery. However they can also be used for malicious activities (e.g., spying or attacks). Similarly, vehicles can be used for benign or nefarious purposes. Thus, detecting drones and vehicles is important to keep public spaces like airports, stadiums, and campuses safe.

Current drone detection systems often use radio communication signals [1, 2] or sound [3]. The first type of system does not work on autonomous drones or those controlled via fiber optics; the second type of system is limited in range. Computer vision, on the other hand, does not have these limitations. For vehicle detection, many detection systems use RGB cameras, which do not perform well at night. Infrared (IR) cameras, on the other hand, operate in day or night. For field-deployable systems, embedded computers are especially useful because they use less size, weight, power and cost (SWaP-C) than desktop computers. This means they can run for longer periods of time off batteries, and are less expensive to deploy to multiple locations for greater coverage.

To prototype a field-deployable, man-portable drone/vehicle detection system, we combine a rugged infrared camera, transformer-based computer vision models, an embedded GPU, and a battery. Our system block diagram and prototype is in Fig. 1. All components run on battery, making the system capable of remote installation. Furthermore the size and weight of our system allows it to be easy to carry.

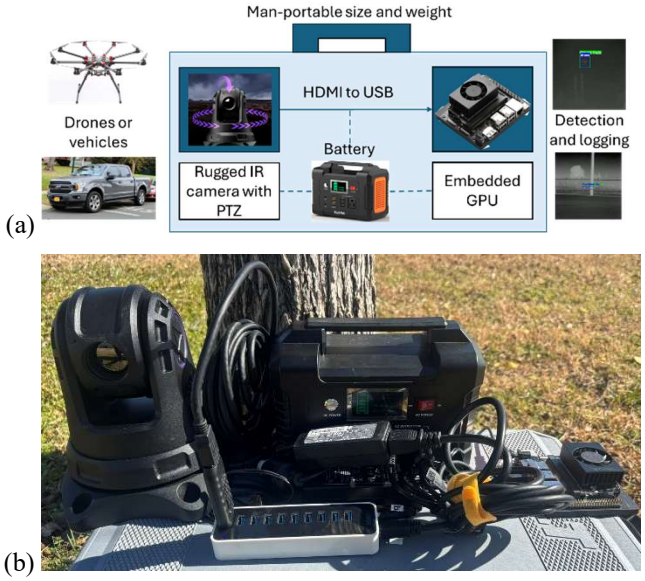

Fig. 1. (a) System block diagram including a rugged infrared camera with pan-tilt-zoom capabilties (middle left), an embedded GPU, (middle right) and a battery (middle). (b) Prototype system. It can be remotely deployed and operate in day or night. Size and weight also allow it be man-portable.

In addition to drone/vehicle detection, our system can adapted to other applications such as detecting pedestrians for surveillance purposes or animals for wildlife monitoring.

II. Related Work

There are existing datasets for drone e.g., [4, 5, 6] and vehicle e.g., [7] computer vision applications. However, most datasets focus on tracking, not detection. Tracking assumes detection has already occurred.

Limited information is given on the cameras used for existing datasets. [4] gives thermal videos at 640 by 512 pixels at 20 frames per second (fps) but no other information is given. The work of [5] has thermal images of 640 x 512 at 25 fps. [6] uses images from an RGB camera which is unlikely to work well at night. [7] has thermal images of different sizes such as 340 x 297, 664 x 419, etc. but the camera types are not given. Because specific camera information is not documented, replicating the hardware setups by other groups is not possible. Knowing just the pixel pitch and focal length of the cameras, e.g., would be useful. That information can be used to calculate the detection, recognition and identification ranges of the Johnson criteria [8]. We perform this estimate in section III.A. (The Johnson criteria defines detection as knowing something is there, recognition as knowing its class type such as drone or bird, and identification as knowing the specific type such as octocopter or seagull.)

There are computer vision models for drone and vehicle detection. As of the writing of this paper, code from [4, 5] did not fully work hence we were not able to use it for benchmarking. We have contacted the authors. [6] uses models for RGB images so comparison with our work is not possible. In [7], they do not develop a model but use common one- and two-stage detectors from 2022. Newer, transformer-based models were not used which are been shown to outperform non-transformer-based models on standard datasets e.g., [9, 10]. In addition, models in the literature are too large for embedded GPUs, hence they are not suited for field-deployable systems.

With regards to hardware, GPU information is not often given [4, 6]. When it is given, desktop GPUs are used, which are not good choices for field-deployable systems. ([7] uses NVIDIA A4000s; [5] uses NVIDIA RTX 2080 Ti's).

III. Methods and Materials

A. Datasets, infrared camera and system components

We recorded two initial datasets, one of a DJI S1000 octocopter drone and another of vehicles, using a Dark 30 infrared camera. In the videos, the drone takes off, flies up and away from the camera, loiters in two separate spots, and then returns to the ground. Behind the drone, power lines and tress are visible; cars and trucks pass-by on a nearby road. For the vehicle dataset, cars and trucks drive past the camera. We did not have control over what types of vehicles passed by, at what times, speeds, etc. In both datasets, targets were centered and tracked manually via pan-tilt-zoom control. The camera does not currently have programmatic control. The first dataset was collected during the day; the second at night. The location of the first dataset was near the University of Texas Arlington's Research Institute (UTARI), while the second was inside UTARI's Maverick Autonomous Vehicle Research Center (MAVRC) pointed at nearby streets. MAVRC is a $2 million, state-of-the-art outdoor netted drone facility equipped with advanced equipment such as 26 Vicon motion capture cameras, lighting for nighttime use, wireless internet coverage and a 66.9 m^2 (720 $foot^2$) climate controlled control/observation center. It is available for use by arrangement. Please see [11].

Sample images are shown in Fig. 2 and further details are in Table 1. The datasets are challenging because of background clutter, foreground occlusions, panning, tilting and zooming of the camera, motion blur, and objects of different scales, locations and types. The images in the drone dataset are also often low contrast due to a data recording issue which was resolved in the vehicle dataset.

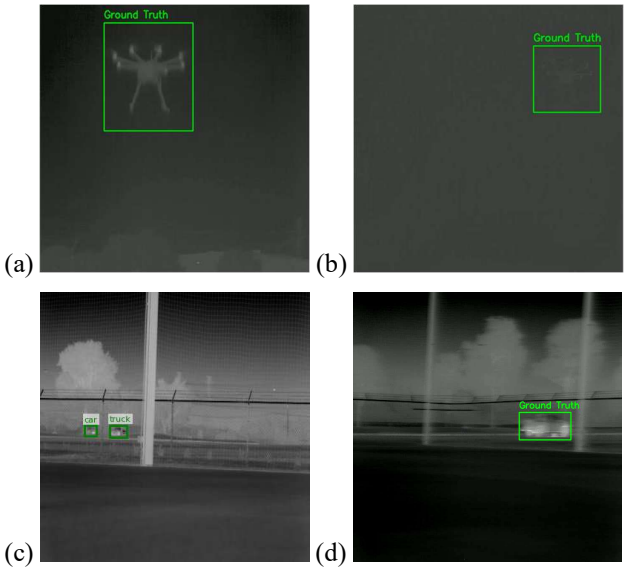

Fig. 2. Sample images from our datasets. (a) and (b) are from the drone dataset; (c) and (d) from vehicles. (a) and (c) are from the training dataset; (b) and (d), testing. Both datasets have clutter, motion blur, etc.

TABLE 1. Infrared datasets

Dataset	Time	Ground truth marked
Drone dataset collected in the day	Training data: 1 minute, 46 seconds (6360 frames) Testing data: 1 mimute, 57 seconds (7020 frames)	5803 frames marked; 4341 had a drone 6408 frames marked; 5953 had a drone
Vehicle dataset collected at night	Training data: 11 seconds (660 frames) Testing data: 56 seconds (3360 frames)	233 frames marked; 23 had car; 148 had truck 1097 frames marked; 481 had car, 351 had truck

Labeling was performed by hand. For the drone dataset, only the drone label was used. For the vehicle dataset, cars were marked as a different category than trucks. Minivans and SUVs were labeled as cars.

Our datasets are small. However that was done to test the ability to fine tune transformer models with limited data. We continue to record and annotate additional data.

The training video was fully used. Validation and testing was performed from the testing videos. There is no overlap between the training and testing videos.

The main preprocessing and data augmentations steps are listed in Table 2. Data augmentations help provide robustness to computer vision models during testing.

TABLE 2. DATA AUGMENTATIONS

Step	Parameters
Stretch image	640x640 pixels
Number of augmented versions of a training image that are generated	2
Rotation	-15° and +15°
Shear	±10° Horizontal, ±10° Vertical
Brightness (constant shift)	Between -15% and +15%
Exposure (gamma correction)	Between -10% and +10%
Blur (Gaussian blur)	Up to 2.5 pixels
Noise (salt and pepper noise)	Up to 0.1% of pixels

For our prototype system, we collected data using a rugged Dark 30 infrared camera with manual pan-tilt and zoom control. Specifications of the camera are listed in Table 3. Full details are available at [12]. The camera is commercially available and thus other groups may replicate our camera setup.

TABLE 3. DARK 30 INFRARED CAMERA SPECIFICATIONS

Specification	Value
Resolution and speed	640x480; 60 fps
Pixel size and spectral range	17x17 µm; 8-14 µm (long wave infrared, a.k.a. LWIR)
Detector NEDT (30 Hz, 300K, F/1) and type	<30 mK; microbolometer
Lens focal length, material, and zoom	30 mm, germanium, digital zoom of 1-8x
Field of view, horizontal, vertical and diagonal	20, 15 and 25 degrees in that order
Size, weight, power and cost	1689 w x 1951 d x 1981 h in mm; 3.15 kg; 9 W; $6500

Because we know the pixel pitch and focal length of the camera, we can calculate the Johnson criteria for detection, recognition and identification for octocopter drones (DJI S1000-sized) and vehicles. These are listed in Table 4 and use the tools of [13] and [14]. ([13] gives ranges for LWIR cameras for vehicles but not drones. [14] gives ranges for RGB cameras for vehicles and drones. To translate the drone ranges from RGB to LWIR, we divide LWIR vehicle ranges by RGB vehicle ranges. These scale factors were then applied to the RGB drones ranges.) In prior datasets, detailed camera information was not given so it is not possible to perform these calculations.

TABLE 4. JOHNSON CRITERIA FOR DARK 30 CAMERA FOR OCTOCOPTERS DRONES (DJI-S1000 SIZE) AND VEHICLES

Target type	Detection range (m)	Recognition range (m)	Identification range (m)
Drones	480	120	74
Vehicles	2029	507	317

We show a picture of our system components in Fig. 1b and estimate its size, weight, power and cost (SWaP-C) in Table 5.

TABLE 5. SIZE, WEIGHT, POWER AND COST (SWAP-C) ESTIMATES FOR OUR SYSTEM COMPONENTS

Component	Size, weight, power and cost (SWaP-C)
Dark 30 infrared camera	Please see Table 3.
HDMI to USB converter	80.77 w x 11.94 d x 30.99 h in mm; 0.02 kg; ~2.5 W; $100
USB hub*	110 w x 43 d x 23 h in mm; 0.22 kg; 10.8 W; $50
NVIDIA Jetson GPU	Please see Table 7.
Portable battery (Flashfish E200)	2083 w x 1575 d x 1092 h in mm; 1.85 kg; -- (151 Wh capacity); $87
Total with 20% margin for wiring, etc.	4650 w x 2341.2 d x 2377.2 h in mm; 6.50 kg; 56.8 W; $8384.40

* If the HDMI to USB converter is plugged directly into the Jetson, the hub is not needed. However it is useful to plug in a keyboard and mouse.

We estimate battery running time for the system at 2 hours, 40 minutes. Additional time can be achieved with additional or larger batteries or possibly solar or wind power.

B. *Computer vision models*

Transformer-based architectures are state-of-the-art for most computer vision tasks. Their core advantage comes from self-attention, which allows the models to capture long-range spatial relationships that convolutional neural networks (CNNs) tend to miss. In addition, they start with a massive "foundation" of knowledge from pre-training on thousands of images, allowing them to adapt to a specific task using only a fraction of the data compared to older deep learning architectures.

Because our datasets are small, this is a good test of the ability of transformer models to do fine tuning with limited data. We chose two recent transformer-based models: YOLOv12 [9] and Roboflow DETR [10]. (We also tried DETR [15] and YOLOv26 [16] but found results to be inconsistent so we omit those results.) YOLOv12 matches the speed of previous CNN-based models while harnessing the performance benefits of area attention and a residual efficient layer aggregation network (R-ELAN). Roboflow DETR uses a DINOv2 backbone and a windowed multi-scale encoder and deformable attention to handle high-resolution inputs without the quadratic memory cost of standard transformers. Interestingly, it no longer uses non-maximum suppression (NMS).

YOLOv12 in its medium size can be run on desktop and embedded GPUs. For the latter, the translation steps are PyTorch->ONNX->TensorRT. This was done for FP32, FP16 and INT8 via Post Quantization Training (PQT) to see what effect model quantization would have. The Roboflow DETR model was only run on a desktop GPU. We are investigating adapting this model for NVIDIA Jetson compatibility.

We performed cross-modal transfer learning. Models were originally trained on RGB datasets such as MS COCO but were fine-tuned on infrared images in our datasets. Although the sensors are quite different, the visual features in RGB images of grass fields, roads, trees, poles, etc. look similar in both images. Thus features learned from the RGB dataset can be fine-tuned on monochrome infrared images.

The primary training parameters are in Table 6. A classification was correct if the model had a confidence score and intersection of union (IoU) both over 50%. Classification metrics of F1, recall, precision, accuracy, and mAP50-95 were measured. Frames per second during testing, excluding image loading and logging, was also recorded to assess speed.

TABLE 6. TRAINING PARAMETERS

Parameter	Value

Maximum epochs with early stopping with patience and warm up epochs	100; 50 for patience; 3 for warm up
Learning rate with momentum	0.01; 0.937
Weight decay	0.0005
Gradient acculumation step	Yes
Learning rule	SGD with momentum

C. GPUs

One of our goals is field-deployable systems thus we developed models on desktop and embedded GPUs. For the desktop GPU, we used a NVIDIA RTX 4070. For the embedded GPU, we used a NVIDIA Jetson Orin Nano. Detailed specifications are in Table 7. Embedded GPUs have less compute capabilities than desktop GPUs but also significantly less size, weight, power and cost (SWaP-C). This makes better suited for field-deployable systems. The embedded GPU offers approximately 3x savings in size, 7x in weight, 9x in power, and more than 2.5x in cost.

TABLE 7. DESKTOP AND EMBEDDED GPU SPECIFICATIONS. FOR SIZE, WEIGHT, POWER, UNDER EMBEDDED, WE LIST SPECIFICATIONS FOR THE DEVELOPMENT BOARD, WHICH HAS A GPU, A CPU AND MORE.

GPU-type	Compute specifications	Size, weight, power and cost (SWaP-C)
Embeded, NVIDIA Jetson Orin Nano (released March 2023)	• GPU: NVIDIA Amphere architecture with 1024 CUDA cores + 32 Tensor cores at max. clock frequency of 1020 MHz • RAM: 8 GB 128-bit LPDDR5 DRAM shared with CPU • CPU: 6-core ARM Cortex-A78AE at a max. clock frequency of 1.7 GHz	• Size: 103 w x 90.5 d x 34.8 h in mm • Weight: 0.17 kg • Power: 7-25 W • Cost: $250
Desktop, NVIDA RTX 4070 (released April 2023)	• GPU: NVIDIA Lovelace architecture with 7680 CUDA cores + 184 Tensor cores + 46 RT cores at max. clock frequency of 2.5 GHz • RAM: 12 GB 192-bit GDDR6X DRAM with max. clock frequency of 1.3 GHz	• Size: 240 w x 40 d x 110 h in mm • Weight: 1.15 kg • Power: 220 W • Cost: $599

IV. RESULTS

A. Drone dataset

For the drone dataset, in Fig. 3 we show several correct results and some incorrect results for Roboflow DETR, the most accurate model. Looking at sample results, the models are able to recognize drones in a variety of scales (e.g., compare Fig. 3a and 3b) and even with low contrast (Fig. 3c and 3d). However there are occasional false negatives (Fig. 3e) and/or false alarms (Fig. 3f).

We list classification and speed metrics in Table 8. With regards to classification metrics, the F1 score, precision, recall, accuracy and mAP50-95 are encouraging but there is certainly room for improvement. Roboflow DETR is the most accurate at 77.8% followed by YOLOv12 on desktop at 66.91%. There is an accuracy drop for YOLOv12 run on desktop v embedded precision options of FP32, FP16 and INT8 as is expected (e.g., desktop 66.91% v embedded FP32 61.42%). INT8 was significantly less accurate than other models. For speed, YOLOv12 is slightly faster than Roboflow DETR (76.2 fps v. 74.1 fps). Speed is reduced from YOLOv12 on the desktop v embedded GPU (e.g., 76.2 fps vs 24.1 fps on Jetson with FP16) but this is also expected. Across the reduced precision runs, one would expect the following order for speed from fastest to slowest: INT8, FP16 and FP32. In our case INT8 had about the same speed as FP16 likely because we are not using NVIDIA's Deep Learning Accelerator (DLA). FP16 is faster than FP32.

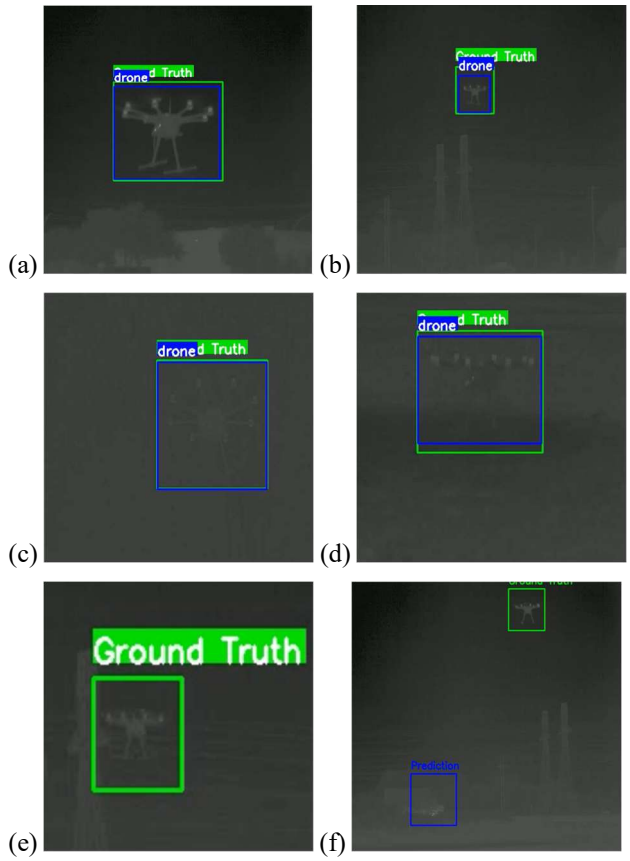

Fig. 3. Sample test results from the drone dataset. (a) through (c) are correct detections; (d) and (e) are not correct (false negative and false psotive in that order).

TABLE 8. DRONE MODEL PERFORMANCE. CLASSIFICATION METRICS ARE F1, PRECISION, RECALL, ACCURACY AND MAP50-95. SPEED IS IN FRAMES PER SECOND.

GPU	YOLOv12	Roboflow DETR
Embedded, INT8	• Classification: 0.51, 1.00, 0.34, 38.66%, 0.46 • Speed: 23.5	--
Embedded, FP16	• Classification: 0.74, 1.00, 0.58, 61.36%, 0.55 • Speed: 24.1 fps	--
Embedded, FP32	• Classification: 0.74, 1.00, 0.58, 61.42%, 0.55 • Speed: 18.6 fps	--
Desktop	• Classification: 0.74, 0.96, 0.60, 66.91%, 0.54 • Speed: 76.2 fps	• Classification: 0.86, 0.87, 0.84, 77.8%, 0.62 • Speed: 74.1 fps

B. *Vehicle dataset*

For the vehicle dataset, in Fig. 4, we show several correct results and some incorrect results for Roboflow DETR, the most accurate model. Examining sample outputs, the models successfully detect vehicles across a range of scales (see Fig. 4a and 4b) and even with significant occlusion (Fig. 4c) and motion blur (Fig. 4d). However, occasional false negatives (Fig. 4e) and false positives (Fig. 4f) are still present.

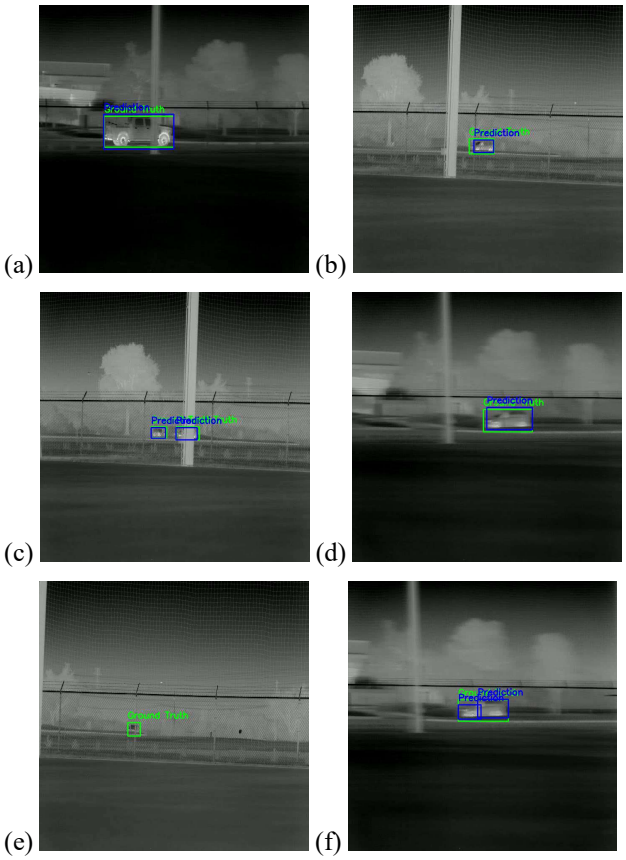

Fig. 4. Sample test results from the vehicle dataset. (a) through (d) are correct detections; (e) and (f) are not correct. (false negative and false psotive in that order).

We list classification and speed metrics in Table 9. As with the drone datasets, classification metrics are promising, though there remains need for refinement. Like the drone dataset, accuracy is best for Roboflow DETR, then YOLOv12 (87.05% v. 61.38%). Comparing desktop v. embedded GPUs, the performance pattern is the same as with drones: performance decreases as precision does and INT8 has a signficant performance drop. (In INT8 however precision increases a small amount.) For speed, as with the drone dataset, YOLOv12 and Roboflow DETR are about the same speed. The rank order of speed on embedded precision models is about the same as for drones: FP32 is the slowest, and FP16 and INT8 are about the same speed. For vehicles, however, INT8 is slightly faster than FP16.

TABLE 9. VEHICLE MODEL PERFORMANCE. CLASSIFICATION METRICS ARE F1, PRECISION, RECALL, ACCURACY AND MAP50-95. SPEED IS IN FRAMES PER SECOND.

GPU	YOLOv12	Roboflow DETR
Embedded, INT8	• Classification: 0.10, 1.00, 0.05, 27.91%, 0.42 • Speed: 23.7 fps	--
Embedded, FP16	• Classification: 0.66, 0.96, 0.51, 61.21%, 0.50 • Speed: 21.9 fps	--
Embedded, FP32	• Classification: 0.66, 0.96, 0.51, 61.21%, 0.50 • Speed: 13.1 fps	•
Desktop	• Classification: 0.67, 0.96, 0.51, 61.38%, 0.49 • Speed: 75.2 fps	• Classification: 0.91, 0.93, 0.89, 87.05%, 0.65 • Speed: 75.4 fps

V. DISCUSSION AND CONCLUSIONS

A. *Summary*

We have shown the components of a field-deployable, man-portable infrared target detection system using a rugged infrared camera, transformer-based computer vision models, an embedded GPU and a battery—please see Fig. 1. Unlike prior work e.g., [4, 5, 7], we consider size, weight, power and cost (SWaP-C) constraints for system components.

For both datasets, models are able to identify targets in a variety of conditions—daytime v nighttime, near v far, in the presence of background clutter and occlusions, with low contrast, etc. This is especially impressive given the small dataset sizes and cross-modal fine tuning. Looking at sample detections, in the case of drone dataset, for example, Fig. 3e shows the system being confused by powerlines in the background. Additional training examples is likely to help here. For the vehicle dataset, Fig. 4f shows the front and rear of a car being recognized as two separate cars. Non-maximal suppression would help overcome this issue; it is not present in Roboflow DETR.

In both cases, accuracy results were encouraging but further gains are possible. Accuracy was highest for Roboflow DETR, then YOLOv12, medium, likely because Roboflow DETR is a larger, newer model. When comparing desktop v. embedded accuracy results, one expects performance to be proportional to precision as we also experienced. The significant performance drops for INT8 might be helped with quantization aware training (QAT), and not just post quantization training (PQT). Across both model type, precision is greater than recall and can this can be tuned by changing the confidence or intersection over union (IoU) thresholds. In terms of speed, YOLOv12 and Roboflow DETR are approximately the same speed. When comparing desktop v. embedded performance results, one expects the desktop to be fastest and then greater speed with less precision, as we experienced with the exception of INT8. For INT8, speeds were similar to FP16. This is likely because we are not using NVIDIA's DLA at the moment. Desktop GPUs however are not good for field deployment. We achieve reasonable speeds on embedded GPUs.

When comparing across datasets, the vehicle dataset appears to be more difficult than the drone dataset because overall performance across models drops slightly. This is likely because the vehicle dataset had two classes while the drone dataset had only one. Differences in speed are likely not significant.

When comparing across GPUs, the embedded GPU had reduced accuracy and speed as is typical. However embedded

GPUs have much less SWaP-C than desktop GPUs and that is better for field deployed system. We obtain competitive inference speeds on embedded GPUs.

B. Additional future directions

1) Datasets, infrared camera and system components

With regards to datasets, benchmarking on standard datasets such as [4, 5, 7] is of interest. However, as stated in section II, standard datasets have limitations of focusing on tracking, not detection; using large computer vision models not suitable for embedded computers; and replicability issues. There are infrared datasets for pedestrians as well [7, 17], and this is an avenue for exploration.

We have shown classification of vehicles into two categories (car v. truck) but doing multiclass classification with additional vehicle (e.g., 18-wheel truck, delivery/flat panel truck, etc.) or drone (e.g., octocopter, hexacopter, quadcopter, etc.) classes is possible. We continue to record and annotate additional data. In addition, looking at tracking, not just detection is of interest.

Our system components would need upgrades to make them truly field deployable. Although the camera is IP 66, rugged batteries, Jetson computers and cables can be used. One can also increase battery life by using larger or more batteries, or integrating solar panels. Although rugged components are larger, heavier and costlier, the system would remain man-portable. Consideration for extreme heat or cold is another consideration.

Lastly, currently pan-tilt-zoom control was done manually. Exploring pan-tilt-zoom control via automatic or semi-automatic methods would especially be useful for remote field operations (e.g., [18]). Other system integration topics include integrating the cameras, embedded computers, and other elements with more rugged components, and communicating detections to central command centers, e.g., via Android TAK. Sensor fusion or cueing with other modalities such as acoustic sensing or radar would also increase detection performance, though with increased system complexity and cost.

2) Computer vision models

Looking at computer vision models, our goal was to test models compatible with embedded GPUs, rather than develop novel models as is common in the literature e.g., [4, 5]. Hyperparameter tuning—both for the model and the hardware—would definitely increase performance. For example, the work of [19] showed that hyperparameter tuning over parameters such as learning rate, batch size and network size can improve performance significantly. For this, several software libraries can be used such as Weights and Biases, Optuna, scikit-learn, etc.

3) GPUs and other computing options

For the hardware further study is possible. Post training quantization (PTQ) and quantization aware training (QAT) are necessary to maximize performance for GPU models [20]. This has also been shown for neuromorphic computing ASICs [19]. In both cases, classification performance and speed drops compared to full precision GPU models but power efficiency and size, weight, power and cost (SWaP-C) are drastically improved. Furthermore, while NVIDIA Jetson embedded GPUs are a popular choice, there are other choices. For example, the Texas Instruments TDA4VM and its starter kit (SK). The TDA4VM chip has a CPU, matrix multiply accelerator (MMA; similar to the NVIDIA's Deep Learning Accelerator or DLA), digital signal processing (DSP), image signal processing (ISP) blocks, etc. Others have reported greater speed [21] and power efficiency compared to embedded GPUs [22].

Still another option for embedded computers include neuromorphic computing. [23] explored neuromorphic ASICs—the BrainChip Akida 1000 (via software simulation) and Intel Loihi 1—for detection and classification problems in the cybersecurity domain. Subsequent work [24] improved classification performance via hyperparameter tuning, and performed experiments on the BrainChip Akida 1000 and Intel Loihi 2 chips. [25] and [26] explored neuromorphic computers for computer vision applications. [25] classified contraband materials across a variety of conditions, such as different temperatures, purities, and backgrounds using simulated long wave infrared (LWIR) hyperspectral imagery. Classification was done using software simulation of a BrainChip Akida 1000. [26] is interesting because they considered neuromorphic cameras as well as neuromorphic computation to track distant small, infrared targets.

We expect future work to advance in this and related application domains across several dimensions.

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